

CSCA Exam (January) Answers and Detailed Analysis

1. Element of Collection and Relation Judgment

- answer : (a)
- Analysis: The set $A=\{6,7,8,9,10\}$, the use of relational symbols between elements and sets, and between sets and sets should be clarified. In (a), 9 is an element of the set A , denoted by " $\in$ ", which is correct; In (b), 6 is a set, and the relationship between sets cannot be expressed with " $\in$ ", it should be changed to " $\subset$ ", which is incorrect; In (c), 6 is an element, and the relationship between elements and sets cannot be expressed with " $\subset$ ", it should be changed to " $\in$ ", which is incorrect; In (d), the empty set $\varnothing$ and the set A have a subset relationship, which should be expressed with " $\subset$ " rather than " $\in$ ", which is incorrect.

2. Union of Sets

- answer : (c)
- Analysis: The set $A=\{x \mid -2 \leq x \leq 2\}$, $B=\{x \mid x > 0\}$. The union $A \cup B$ is the set composed of all elements belonging to A or belonging to B . A contains all real numbers from -2 to 2 , B contains all real numbers greater than 0 , and the union covers all real numbers from -2 to $+\infty$, i.e., $x \mid x \geq -2$, corresponding to option (c).

3. Solving Quadratic Inequalities

- answer : (a)
- Analysis: The inequality $x^2 - x - 2 > 0$, factorization first: $(x - 2)(x + 1) > 0$. Quadratic function $y = x^2 - x - 2$ opens upward, with zeros at $x = -1$ and $x = 2$. The upward-opening parabola has the part greater than 0 outside the two zeros, i.e., $x < -1$ or $x > 2$, corresponding to option (a).

4. Application of General Term Formula of Arithmetic Sequence

- answer : (a)
- Analysis: The general term formula of an arithmetic sequence is $a_n = a_1 + (n - 1)d$, where $a_1 = 2$ (first term), $d = 3$ (common difference), $n = 100$. Substitute for calculation: $a_{100} = 2 + (100 - 1) \times 3 = 2 + 99 \times 3 = 2 + 297 = 299$, corresponding to option (a).

5. Function Domain Solving

- answer : (c)
- Analysis: Function $f(x) = \frac{1}{x} + \sqrt{1 - x}$, The domain must satisfy the meaning of each part.
Fraction $\frac{1}{x}$ The required denominator is not 0 , i.e., $x \neq 0$; The square root $1 - x$ requires that the number under the square root is non-negative, that is, $1 - x \geq 0$, we get $x \leq 1$. Combining the two, the domain is $(-\infty, 0) \cup (0, 1]$, corresponding to option (c).

6. Coordinate Characteristics of Points Symmetric About the X-Axis

- answer : (b)
- Analysis: In the Cartesian coordinate system, when two points are symmetric about the x axis, their horizontal coordinates remain unchanged while the vertical coordinates are opposites. The point $P(1,2)$ is symmetric about the x axis, with the horizontal coordinate still being 1 and the vertical coordinate becoming -2 , i.e., $Q(1,-2)$. The corresponding option is (b).

7. Trigonometric Values of Special Angles

- answer : (a)
- Analysis: Given $\angle = 30^\circ$, the trigonometric values of special angles should be memorized:

$$\sin 30^\circ = \frac{1}{2}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}, \quad \tan 30^\circ = \frac{\sqrt{3}}{3}$$

The comparison options are: (a)

correct, and (b), (c), and (d) all incorrect.

8. Odd-Even Function Test

- answer : (d)
- Analysis: The parity judgment requires the domain to be symmetric about the origin, and $f(-x) = f(x)$ (even function) or $f(-x) = -f(x)$ (odd function). The function $f(x) = x^4 + 3$ has a domain of $\mathbb{R}$ (symmetric about the origin). Calculation
 $f(-x) = (-x)^4 + 3 = x^4 + 3 = f(x)$, satisfies the definition of an even function, corresponding to option (d).

9. Inverse Function Solve

- answer : (a)
- Analysis: Find $y = x^3 + 3$ (The inverse function of $x \in \mathbb{R}$) is as follows: Solve the equation about $x: x^3 = y - 3$, get $x = \sqrt[3]{y - 3}$; Swap x and y , we get the inverse function $y = x^3 + 3$. The domain of the original function is $\mathbb{R}$, and the range is $\mathbb{R}$, so the domain of the inverse function is also $\mathbb{R}$, corresponding to option (a).

10. Coordinate Characteristics of Quadrant Points

- answer : (b)
- Analysis: In the Cartesian coordinate system, the coordinates of a point in the third quadrant are both negative. Analysis options: (a) $(-1,2)$ (negative x -coordinate, positive y -coordinate, second quadrant); (b) $(-1,-2)$ (both negative coordinates, third quadrant); (c) $(1,2)$ (first quadrant); (d) $(1,-2)$ (fourth quadrant), corresponding to option (b).

11. Monotonicity of Absolute Value Function

- answer : (b)
- Analysis: The function $y = |x|$ can be expressed in segments as $y = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$. Monotonicity analysis: When $x > 0$, $y = x$ is an increasing function; When $x < 0$, $y = -x$ is a decreasing function; On

the entire domain $\mathbb{R}$, the function is not monotonic (decreasing first, then increasing). Therefore

(a) (c) (d) Wrong, (b) Correct.

12. Fractional Inequality Solution

- answer : (a)
- Analysis: Inequality $\frac{2x+1}{x-2} \leq 0$, Fractional inequalities need to be converted to integer inequalities (note that the denominator is not 0):

$(2x+1)(x-2) \leq 0$ and $x \neq 2$. After factorization, the zeros are $x = -\frac{1}{2}$. And $x=2$, the quadratic function opens upward, the part less than or equal to 0 is between the two zeros, combined with the denominator is not 0, the solution set is $[-\frac{1}{2}, 2)$, corresponding to option (a).

13. Arithmetic Mean Calculation

- answer : (b)
- Analysis: First calculate the value of a, b , $a=2-3, b=2+3$, using the square difference formula:

$$ab = (2-3)(2+3) = 2^2 - (3)^2 = 4 - 9 = -5. \text{ The arithmetic mean is } \frac{a+b}{2} \text{ Calculation } a+b =$$

$$(2-3) + (2+3) = 4, \text{ so the arithmetic mean is } \frac{4}{2} = 2, \text{ Select option (b).}$$

14. Slope Calculation

- answer : (c)
- Analysis: A line passing through two points $A(x_1, y_1)$ and $B(x_2, y_2)$ has a slope formula of

$$k = \frac{y_2 - y_1}{x_2 - x_1}. \text{ Substituting } A(-2, 3) \text{ and } B(3, 1), \text{ we get } k = \frac{1-3}{3-(-2)} = \frac{-2}{5} = -\frac{2}{5},$$

Select option (c).

15. Trigonometric Values of Right Angles

- answer : (b)
- Analysis: The line passing through the point $P(1, 3)$ forms an angle α with the x axis, then the slope of

the line $k = \tan \alpha = \frac{3-0}{1-0} = 3$. Construct a right triangle with the opposite side being 3, the adjacent side being 1, and the hypotenuse being

$$\sqrt{1^2 + 3^2} = \sqrt{10}. \sin \alpha = \frac{\text{对边}}{\text{斜边}} = \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}, \text{ Select option (b).}$$

16. Arithmetic Sequence Test

- answer : (a)
- Analysis: An arithmetic sequence is defined as a sequence where the difference between any two consecutive terms is a constant (common difference). ① 7, 13, 19, 25: The difference $13 - 7 = 6$, is an arithmetic sequence; ② 2, 4, 7, 11: $4 - 2 = 2, 7 - 4 = 3$, the differences are not equal, it is not an arithmetic sequence; ③ -1, -3, -5, -7: The difference $-3 - (-1) = -2$, is an arithmetic sequence; ④ 2, 4, 8, 16: is a geometric sequence (common ratio 2), not an arithmetic sequence. In summary, there are 2 arithmetic

sequences, corresponding to option (a).

17. Distance Formula Between Two Points

- answer : (b)

- Analysis: The distance formula between two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is $|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Substitute $A(3, -2)$ and $B(-5, -1)$, 得
 $|AB| = \sqrt{(-5 - 3)^2 + (-1 - (-2))^2} = \sqrt{(-8)^2 + 1^2} = \sqrt{64 + 1} = \sqrt{65}$, Option (b).

18. Solve Linear Equation (slope + Point)

- answer : (b)
- Analysis: The slope of a line with an inclination angle of 45° is $k = \tan 45^\circ = 1$. The line passes through the point $(0, 2)$ and can be expressed in the intercept form as $y = kx + b$. Since $b = 2$, the equation of the line is $y = x + 2$, which corresponds to option (b).

19. The General Equation of a Circle Is Transformed Into the Standard Form.

- answer : (c)
- Analysis: The general equation of a circle is $x^2 + y^2 - 4x - 3 = 0$. By completing the square, it can be transformed into the standard form.
 $(x - h)^2 + (y - k)^2 = r^2$ ((h, k) with the center as the origin, r as the radius). Solution steps: $x^2 - 4x + y^2 = 3$, $(x^2 - 4x + 4) + y^2 = 3 + 4$, i.e. $(x - 2)^2 + y^2 = 7$. Therefore, the center is $(2, 0)$ and the radius is $\sqrt{7}$.
 Corresponding option is (c).

20. Trigonometric Relationships

- answer : (b)
- Analysis: Known $\sin \alpha = \frac{4}{5}$, α in the second quadrant (second quadrant $\cos \alpha < 0$). By the fundamental trigonometric relationship $\sin^2 \alpha + \cos^2 \alpha = 1$, we get
 $\cos \alpha = -\sqrt{1 - \sin^2 \alpha} = -\sqrt{1 - \left(\frac{4}{5}\right)^2} = -\sqrt{\frac{9}{25}} = -\frac{3}{5}$.
 $\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{\frac{4}{5}}{-\frac{3}{5}} = -\frac{4}{3}$, Hence (a) (c) (d) is wrong, (b) is correct.

21. Solving the First Term and Common Difference of an Arithmetic Sequence

- answer : (a)
- Analysis: The general term formula of an arithmetic sequence is $a_n = a_1 + (n - 1)d$. Given that $a_2 = 1$ and $a_4 = 5$, set up a system of equations: $\begin{cases} a_1 + d = 1 \\ a_1 + 3d = 5 \end{cases}$. Subtract the first equation from the second: $2d = 4$, get $d = 2$, substitute into the first equation to get $a_1 = 1 - 2 = -1$.
 Therefore, $a_1 = -1, d = 2$, corresponding to option (a).

22. Inequality Property Judgment

- answer : (b)
- Analysis: Analyze each option: (a) If $a > b$, when $c \leq 0, ac > bc$ does not hold (e.g., when $c = 0, ac = bc$); (b) If $a > b$, adding c to both sides without changing the direction of the inequality, i.e. $a + c > b + c$, always holds;

(c) If $ac > bc$, when $c < 0$

At this point, the direction of the inequality needs to be reversed, i.e., $a < b$, which does not hold; (d) If $a > b$, such as $a = 1, b = -2$, then $a^2 = 1 < b^2 = 4$, which does not hold. Therefore, (b) is correct.

23. Exponential Function Property Judgment

- answer : (c)
- Analysis: Exponential function $y=a^x$ ($a>0$ and $a\neq 1$) properties: (a) The range is $(0, +\infty)$, correct; (b) When $x=0, y=a^0=1$, the graph passes through $(0,1)$, correct; (c) Monotonicity is determined by a : increasing function when $a>1$, decreasing function when $0<a<1$, "increasing function on the domain" is incorrect; (d) The domain is $\mathbb{R}$, correct. Therefore (c) is the incorrect option.

24. Application of Double Angle Sine Formula

- answer : (b)
- Analysis: Given that $\sin \alpha = \frac{3}{5}$ α in the first quadrant ($\cos \alpha > 0$). By the trigonometric relationship of the same angle $\cos \alpha = \sqrt{1 - \sin^2 \alpha} = \sqrt{1 - (\frac{3}{5})^2} = \frac{4}{5}$. The formula for the sine of the double angle $\sin 2\alpha = 2 \sin \alpha \cos \alpha$, substituting it gives $\sin 2\alpha = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$, Select option (b).

25. Solve the Intersection of Two Lines

- answer : (d)
- Analysis: Find the line $l_1: 3x - y + 8 = 0$ and $l_2: x + 2y - 9 = 0$ intersection points, solve the system of equations:
$$\begin{cases} 3x - y = -8 \\ x + 2y = 9 \end{cases}$$
 From the first equation, we get $y = 3x + 8$. Substitute into the second equation: $x + 2(3x + 8) = 9, x + 6x + 16 = 9, 7x = -7, x = -1$. Then $y = 3 \times (-1) + 8 = 5$. The intersection coordinates are $(-1, 5)$, corresponding to option (d).

26. Hyperbolic Property Analysis

- answer : (a)
- Solution (analysis): Double curve, real semi-axis of linear equation $\frac{x^2}{64} - \frac{y^2}{16} = 1$, $b^2 = 16$, $a^2 = 64$, $c^2 = a^2 + b^2 = 80, c = 4\sqrt{5}$, hence the focal length $2c = 8\sqrt{5}$; eccentricity $e = \frac{c}{a} = \frac{4\sqrt{5}}{8} = \frac{\sqrt{5}}{2}$. Analysis options: (a) Correct, (b) (c) (d) Incorrect.

27. Application of Monotonicity of Exponential Function and Power Function

- answer : (d)
- Analysis: Using the monotonicity of functions to determine inequalities: (a) Power functions

$y = x^{-2}$ (即 $\frac{1}{x^2}$), is monotonic decreasing on $(0, +\infty)$, $2.1 > 1.2$, so $2.1^{-2} < 1.2^{-2}$, wrong; (b) the exponential function $y = 0.75^x$ ($0 < 0.75 < 1$) is monotonic decreasing, $-0.2 > -0.4$, so $0.75^{-0.2} < 0.75^{-0.4}$, wrong; (c) the exponential function $y = 3^x$ ($3 > 1$) monotone increasing, $2.25 < 3$, hence $3^{2.25} < 3^3$, error; (d) power function $y = x^{\frac{2}{3}}$ (i.e. $x^{\frac{2}{3}}$), is monotonically increasing on $(0, +\infty)$, $2.1 > 1.2$, 故 $2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$, exactness.

28. Standard Equation of a Circle

- answer : (d)
- Analysis: The standard equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle, and r is the radius. Center $(-3, 2)$, radius 4, substituting yields $(x - (-3))^2 + (y - 2)^2 = 4^2$, i.e. $(x + 3)^2 + (y - 2)^2 = 16$, corresponding option (d).

29. Logarithm Rule of Operation Judgment

- answer : (b)
- Analysis: Logarithmic operation rules ($a > 0, b > 0, a \neq 1, b \neq 1$): (a) $\ln(ab) = \ln a + \ln b$, correct; (b) the formula for base change is $\log_a b = \frac{\ln b}{\ln a}$, rather than $\frac{\ln a}{\ln b}$, error; (c) $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$, Correct; (d) $\ln(ab) = \ln a + \ln b$, which is correct. Therefore, (b) is incorrect.

30. Properties of Sinusoidal Function

- answer : (c)
- Analysis: Properties of the sine function $y = \sin x$: (a) Maximum value 1, minimum value -1 , correct; (b) Period is 2π , it is a periodic function, correct; (c) Odd-even property: $\sin(-x) = -\sin x$, it is an odd function, not an even function, incorrect; (d) Domain is $\mathbb{R}$, correct. Therefore, (c) is the incorrect option.

31. Formula of Focal Radius of Parabola

- answer : (d)
- Analysis: The parabola $y^2 = 4x$, standard form $y^2 = 2px$ ($p = 2$), focus $F\left(\frac{p}{2}, 0\right) = (1, 0)$. The formula for the radius of a parabola (the distance from a point $P(x_0, y_0)$ on the parabola to the focus): $|PF| = x_0 + \frac{p}{2}$. If the P coordinate $x_0 = 4$, then $|PF| = 4 + 1 = 5$, corresponding to option (d).

32. Conditions for Two Lines to Be Perpendicular

- answer : (d)
- Analysis: Two lines $l_1 : A_1x + B_1y + C_1 = 0$ and $l_2 : A_2x + B_2y + C_2 = 0$ are perpendicular if $A_1A_2 + B_1B_2 = 0$. Substituting $l_1: ax + (a - 1)y + 3 = 0$ and $l_2: 2x + ay - 1 = 0$, we get $a \times 2 + (a - 1) \times a = 0$, rearranged: $2a + a^2 - a = 0, a^2 + a = 0, a(a + 1) = 0$, solving for $a = 0$ or $a = -1$, corresponding to option (d).

33. Application of the Distance Formula Between Two Points

- answer : (b)
- Analysis: Two points $P(-1, 2)$ and $Q(3, 1)$, distance formula $|PQ| = \sqrt{(3 - (-1))^2 + (1 - 2)^2} = \sqrt{4^2 + (-1)^2} = \sqrt{16 + 1} = \sqrt{17}$, corresponding to option (b).

34. The Formula for the Sine of the Sum of Two Angles

- answer : (c)
- Analysis: Known $\cos \alpha = -\frac{\sqrt{5}}{5}$, $\alpha \in (\frac{\pi}{2}, \pi)$ ($\sin \alpha > 0$). From the trigonometric relationship of the same angle, we get

$$\sin \alpha = \sqrt{1 - \cos^2 \alpha} = \sqrt{1 - \frac{5}{25}} = \sqrt{\frac{20}{25}} = \frac{2\sqrt{5}}{5}.$$

The sine formula

of the sum of two angles $\sin(A+B) = \sin A \cos B + \cos A \sin B$, then

$$\sin\left(\frac{\pi}{6} + \alpha\right) = \sin \frac{\pi}{6} \cos \alpha + \cos \frac{\pi}{6} \sin \alpha = \frac{1}{2} \times \left(-\frac{\sqrt{5}}{5}\right) + \frac{\sqrt{3}}{2} \times \frac{2\sqrt{5}}{5} = -\frac{\sqrt{5}}{10} + \frac{2\sqrt{15}}{10} =$$

$$\frac{2\sqrt{15} - \sqrt{5}}{10}, \text{ corresponding option (c).}$$

35. Same Function Judgment

- answer : (b)
- Analysis: The same function must satisfy "the same domain and the same correspondence." (a) $y = (x)^2$ (domain $x \geq 0$) and $y = \sqrt{x^2}$ (Domain $\mathbb{R}$), different domains, not the same function; (b) $y = \frac{x^4 - 1}{x^2 + 1}$ (Domain $\mathbb{R}$), simplified to $y = x^2 - 1$ (domain $\mathbb{R}$), the domain and correspondence are identical, making them the same function; (c) $y = 1$ (constant function) and $y = x^2$ (quadratic function), the correspondence differs, making them not the same function; (d) $y = x$ (range $\mathbb{R}$) and $y = \sqrt{x^2} = |x|$ (domain $[0, +\infty)$), the corresponding relationship is different, not the same function. Therefore, (b) is correct.

36. Half Cosine Formula

- answer : (b)
- Analysis: Given $\sin \alpha = -\frac{12}{13}$, $\alpha \in (\pi, \frac{3\pi}{2})$, $\frac{\alpha}{2} \in (\frac{\pi}{2}, \frac{3\pi}{4})$ (Second quadrant, $\cos \frac{\alpha}{2} < 0$).

From the trigonometric relationship of the same angle

$$\cos \alpha = -\sqrt{1 - \sin^2 \alpha} = -\sqrt{1 - \frac{144}{169}} = -\frac{5}{13}.$$

half cosine formula

$$\cos \frac{\alpha}{2} = -\sqrt{\frac{1 + \cos \alpha}{2}} \quad \left(\text{follow } \frac{\alpha}{2} \text{ In the second quadrant, with a}\right.$$

negative sign, substituting yields

$$\cos \frac{\alpha}{2} = -\sqrt{\frac{1 - \frac{5}{13}}{2}} = -\sqrt{\frac{4}{13}} = -\frac{2\sqrt{13}}{13}, \quad \text{Select option}$$

(b).

37. Half-Sinus Formula

- answer : (d)

- Analysis: Known $\cos \alpha = -\frac{1}{2}$, $\alpha \in (\frac{\pi}{2}, \pi)$, then $\frac{\alpha}{2} \in (\frac{\pi}{4}, \frac{\pi}{2})$ (first quadrant, Half-Sin $\sin \frac{\alpha}{2} > 0$) formula $\sin \frac{\alpha}{2} = \sqrt{\frac{1 - \cos \alpha}{2}}$ (因 $\frac{\alpha}{2}$ In the first quadrant, taking the positive value, we obtain

$$\sin \frac{\alpha}{2} = \sqrt{\frac{1 - (-\frac{1}{2})}{2}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2} ? \quad \text{Correction: Calculation error. Recalculate:}$$

$$\frac{1 - \cos \alpha}{2} = \frac{1 - (-\frac{1}{2})}{2} = \frac{\frac{3}{2}}{2} = \frac{3}{4}, \quad \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2} ? \quad \text{But option (b) is } \frac{3}{2}, \quad (d) \text{ yes } \frac{1}{2}. \quad \alpha \in (\frac{\pi}{2}, \pi), \quad \cos \alpha = -\frac{1}{2}, \text{ then } \alpha = \frac{2\pi}{3}, \quad \frac{\alpha}{2} = \frac{\pi}{3}, \quad \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}, \quad \text{Option (b)? Wrong, the correct}$$

answer is (b).

reinspection :

Original analysis has

38. Induced Trigonometric Formulas

- answer : (d)
- Analysis: Core of the Induction Formula: $\sin(\pi + \alpha) = -\sin \alpha$, $\cos(\pi + \alpha) = -\cos \alpha$, $\sin(-\alpha) = -\sin \alpha$, $\cos(-\alpha) = \cos \alpha$. Comparison Options: (a) (b) (c) Wrong, (d) Correct.

39. Properties of the Tangent Function

- answer : (d)
- Analysis: Properties of the tangent function $y = \tan x$: (a) The range is $\mathbb{R}$, not $[-1, 1]$, which is incorrect; (b) The domain is $x | x \neq k\pi + \frac{\pi}{2}, k \in \mathbb{Z}$, non $\mathbb{R}$, error; (c) Odd-even property: $\tan(-x) = -\tan x$, is an odd function, not an even function, error; (d) The minimum positive period is π , correct. Therefore, (d) is correct.

40. Derivation of the General Term Formula of a Sequence

- answer : (b)
- Analysis: Sequence: $-\frac{1}{5}, \frac{1}{7}, -\frac{1}{9}, \frac{1}{11}, \dots$. Observation rules: ① Symbol: Odd terms are negative, even terms are positive, and are expressed by $(-1)^n$; ② Denominator: $5=2 \times 1+3, 7=2 \times 2+3, 9=2 \times 3+3, 11=2 \times 4+3$, i.e. the denominator is $2n+3$; ③ Molecule: Both are 1. In conclusion, the general formula $a_n = \frac{(-1)^n}{2n+3}$, Select option (b).

41. Logarithmic Function Passing Through Fixed Point Problem

- answer : (d)
- Analysis: The logarithmic function $y = \log_a x$ ($a > 0$ and $a \neq 1$) Always passes through the fixed point $(1, 0)$. For the function $f(x) = 2 + \log_a(x-3)$, let $x-3=1$ (i.e., $x=4$), then $\log_a 1 = 0$, so $f(4) = 2+0=2$. Therefore, the coordinates of the fixed point P are $(4, 2)$, and the corresponding option is (d).

42. Parabola of Parabola

- answer : (a)

- Analysis: The parabola $y^2 = -x$, standard form $y^2 = -2px$ ($p = \frac{1}{2}$), the equation of the focus is $x = \frac{p}{2}$.
substitution $p = \frac{1}{2}$ equation of the line of sight $x = \frac{1}{4}$, corresponding option (a).

43. Application of the Definition of Ellipse

- answer : (b)
- Analysis: Definition of an ellipse: The set of points in a plane that are equidistant from two foci F_1 and F_2 such that the sum of the distances from these points to the foci is a constant ($2a, 2a > |F_1 F_2|$). Equation of an ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$, standard form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a^2 = 4$, i.e. $a = 2$).
Hence $|PF_1| + |PF_2| = 2a = 4$, corresponding to option (b).

44. Vector Collinearity Test (Three Points Collinearity)

- answer : (b)
- Analysis: The vector condition for three collinear points: If $\overrightarrow{AB} = k\overrightarrow{AD}$ (where k is a constant), then A, B, D are collinear. Calculation of vectors:
 $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} = (\vec{a} + 5\vec{b}) + (-2\vec{a} + 8\vec{b}) + (3\vec{a} - 3\vec{b}) = 2\vec{a} + 10\vec{b} = 2(\vec{a} + 5\vec{b}) = 2\overrightarrow{AB}$
Therefore, if $\overrightarrow{AD}$ and $\overrightarrow{AB}$ are collinear and have a common point A , then A, B , and D are collinear, and the corresponding option is (b).

45. Solve Linear Equation (perpendicular + Through Intersection Point)

- answer : (a)
- Analysis: Step 1: Find the intersection of $l_2: x + y + 1 = 0$ and $l_3: 2x + y - 1 = 0$, solve the system of equations: $\begin{cases} x + y = -1 \\ 2x + y = 1 \end{cases}$, subtracting the two equations gives $x = 2$, substituting gives $y = -3$, the intersection point is $(2, -3)$. Step 2: Find the slope of line l , l is perpendicular to $l_1: x + 2y + 4 = 0$, l_1 is $-\frac{1}{2}$. Then, the slope of l is 2 (the product of the slopes of vertical lines is -1). Step 3: Use the point-slope formula $y - y_0 = k(x - x_0)$, substitute $(2, -3)$ and $k = 2$, get $y - (-3) = 2(x - 2)$, rearrange to $2x - y - 7 = 0$, corresponding to option (a).

46. The Relationship Between Complex Numbers and the Roots of Quadratic Equations

- answer : (b)
- Analysis: The points corresponding to the complex number z lie on the line $x - y = 0$, so $z = a + ai$ ($a \in \mathbb{R}$). z is the root of the equation $x^2 + mx + 4 = 0$. Substituting yields $(a + ai)^2 + m(a + ai) + 4 = 0$. Expansion:
 $a^2 + 2a^2i - a^2 + ma + mai + 4 = 0$, organizing the real part and imaginary part: $(ma + 4) + (2a^2 + ma)i = 0$. complex
If the number is 0, both the real part and the imaginary part must be 0. Set up the equation system:
 $\begin{cases} ma + 4 = 0 \\ 2a^2 + ma = 0 \end{cases}$. From the first equation, we get $ma = -4$, substituting into the second equation: $2a^2 - 4 = 0$, $a^2 = 2$, $a = \pm \sqrt{2}$. When $a = \sqrt{2}$, $m = -\frac{4}{\sqrt{2}} = -2\sqrt{2}$; When $a = -\sqrt{2}$, $m = \frac{4}{\sqrt{2}} = 2\sqrt{2}$. Therefore, $m = \pm 2\sqrt{2}$, corresponding to option (b).

47. General Term and Sum of the First N Terms of a Sequence

- answer : (c)
- Analysis: The sequence an satisfies $a_1=1, \frac{1}{a_n} + \frac{2}{a_{n+1}} = 0$, Organized $\frac{1}{a_{n+1}} = -\frac{1}{2a_n}$, i.e. $a_{n+1} = -2a_n$. Therefore an is a geometric sequence with the first term $a_1=1$ and common ratio $q=-2$, and the general term $a_n = (-2)^{n-1}$. Sequence $b_n = |a_n| = |(-2)^{n-1}| = 2^{n-1}$, is a geometric sequence with the first term 1 and common ratio 2. The sum of the first n terms $S_n = \frac{1(2^n - 1)}{2 - 1} = 2^n - 1$,

Option (c).

48. Probability calculation (independent events + opposite events)

- answer : (b)
- Analysis: Using the "opposite event" method to simplify calculations, the probability that two people prefer different apps equals 1 - the probability that they prefer the same app. Total population: 200 high school students and 80 college students. Calculating the probability of the same app: ① Both prefer App A: $\frac{80}{200} \times \frac{30}{80} = \frac{3}{20}$; ② Both prefer Application B: $\frac{40}{200} \times \frac{20}{80} = \frac{1}{20}$; ③ They all prefer using C: $\frac{60}{200} \times \frac{20}{80} = \frac{3}{40}$; ④ They all like to use D: $\frac{20}{200} \times \frac{10}{80} = \frac{1}{80}$ Total probability of the same application: $\frac{3}{20} + \frac{1}{20} + \frac{3}{40} + \frac{1}{80} = \frac{12}{80} + \frac{4}{80} + \frac{6}{80} + \frac{1}{80} = \frac{23}{80}$. Therefore, the probability of different applications: $1 - \frac{23}{80} = \frac{57}{80}$, corresponding option (b).